

Pharma Supply Chain Risk Model

This Power BI analytics solution merges regulatory data, chemical informatics, and financial market metrics into a unified predictive framework. By analyzing the structural profiles of small-molecule drugs alongside market volatility and FDA shortage logs, the model identifies systemic vulnerabilities inherent to the pharmaceutical supply chain.

Technical Architecture

Built as a high-performance Tabular Model (Compatibility Level 1600), the project features an optimized star/snowflake schema designed for instantaneous cross-filtering. It relies on a core bridge table (shortage match) to seamlessly link molecular clusters (cluster output) directly with operational logs (fda shortage). All time-series market variables and shortage timelines are synchronized through a centralized calendar dimension (PaddedDateTableDates) to support advanced DAX time-intelligence metrics.

Ultimately, this model shifts pharmaceutical risk management from a reactive state to a proactive, structure-aware discipline, allowing analysts to intercept supply chain vulnerabilities before they impact global healthcare markets.

